

SESHANAGOTTU VENKATA SUJITH PRAVEEN

AI/ML Engineer: RAG & LLM Systems

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Available immediately | Open to full-time and contract roles

SUMMARY

AI/ML Engineer focused on retrieval-augmented generation and LLM applications. Built a hybrid retrieval and multi-agent system during a 3-month AI internship, and independently shipped three public projects: a governed clinical RAG system, a PyPI-published mutation-testing CLI, and a fully client-side job platform. Comfortable across retrieval, evaluation, FastAPI backends and React frontends.

TECHNICAL PROJECTS

RespiRAG: Governed Clinical RAG for Oncology (NSCLC)

github.com/SVSPraveen/RespiRAG

Python, FastAPI, React, TypeScript, Qdrant, MedCPT, PostgreSQL, Redis, Groq, Docker | Live demo: pharma-spine-ai.vercel.app

- Built a citation-grounded RAG pipeline over FDA drug labels and clinical guidelines: HyDE query expansion, MedCPT embeddings, top-20 Qdrant retrieval, then cross-encoder reranking with a 0.15 relevance gate so weak matches return a safe refusal instead of an LLM guess.
- Added an LLM-as-judge verifier that checks each answer against retrieved chunks, and a PASS / DEFLECT / DENY governance layer (jailbreak, prompt-injection and lethal-dose blocking) with every decision logged to a PostgreSQL audit trail.
- Built role-based access (Patient / HCP / Admin) with JWT refresh rotation and Redis revocation, rate limiting, Redis citation caching, and Groq key rotation with an Ollama fallback.
- Wrote an offline evaluation harness (15 golden NSCLC questions, CI-gated at a 75% keyword hit rate) and document version control that flags superseded prescribing information and purges its vectors.

DeployProof: In-Memory AST Mutation-Testing CLI

github.com/SVSPraveen/deployproof

Python, AST, pytest, GitHub Actions, OSV | pypi.org/project/deployproof

- Published a Python CLI (pip install deployproof, v1.1.19, 278 passing tests, 15 GitHub stars) that runs mutation testing scoped to the git diff, exposing test suites with 100% line coverage that verify very little.
- Compiles all mutants into one conditional AST in memory (no disk rewrites); built a self-healing synthesizer that generates pytest cases to kill surviving mutants, plus parallel full-repo sweeps and WSL delegation.
- Added companion gates for OWASP-style SAST, git-history secrets, hallucinated or vulnerable dependencies (PyPI + OSV) and symlink escapes, wired into a pre-push hook and GitHub Actions.

SPrav Job AI: Client-Side Job Search Platform

[SPrav-JOB-AI](https://sprav-job-ai.vercel.app)

React, Vite, WebGPU, IndexedDB, Node.js, Python

- Built a job platform with no backend: pulls live listings from public ATS endpoints (Greenhouse, Lever, Ashby, SmartRecruiters and others) and Hacker News "Who is Hiring", storing user data in an encrypted IndexedDB vault.
- Wrote a standard-library Python harvester on a GitHub Actions cron that enforces a 14-day freshness limit, filters evergreen/ghost postings and de-duplicates by SHA-256 fingerprint.
- Built a 12-dimension ATS scoring engine with vendor-specific rules and a dependency-free in-browser PDF resume compiler with a guardrail that checks generated bullets against verified profile facts.
- Implemented hardware-aware inference: local quantized Qwen 2.5-Coder via WebGPU, falling back to Ollama or six bring-your-own-key cloud providers; covered by 488 automated tests (Node test runner and Vitest) with lint and CSP checks.

EXPERIENCE

AI/ML Engineer Intern, MobcoderAI, Noida (Onsite)

Apr 2026 - Jul 2026

- Built a hybrid BM25 + dense retrieval pipeline with Reciprocal Rank Fusion and cross-encoder reranking on Qdrant Cloud for SEC 10-K/10-Q search across 4 companies; added a semantic cache (cosine similarity ≥ 0.92) that served repeat queries without an LLM call.
- Designed a multi-agent system (6 specialized agents plus a router) routing between Llama 3.1 8B and Llama 3.3 70B on Groq to balance cost, latency and accuracy.
- Built a deterministic, LLM-free evaluation harness (RapidFuzz) scoring faithfulness, groundedness, hallucination rate and citation accuracy, with numerical claims gated to 1% tolerance and tracked in MLflow.

SKILLS

LLM & RAG: RAG, hybrid search (BM25 + dense), HyDE, Reciprocal Rank Fusion, cross-encoder reranking, semantic caching, multi-agent orchestration, prompt engineering

Models & Vector Search: Llama 3.x, Qwen 2.5, Groq, Ollama, OpenAI API, Qdrant, MedCPT, Sentence-Transformers, HuggingFace

Evaluation: Hallucination and citation-accuracy evaluation, MLflow, pytest, AST mutation testing

Backend & Frontend: Python, FastAPI, REST, PostgreSQL, Redis, SQLAlchemy, React, TypeScript, JavaScript

Tools: Docker, Git, GitHub Actions (CI/CD), JWT authentication, Vercel, CloudFlare

EDUCATION

B.Tech, Computer Science & Engineering, Amity University, Uttar Pradesh

2022 - 2026